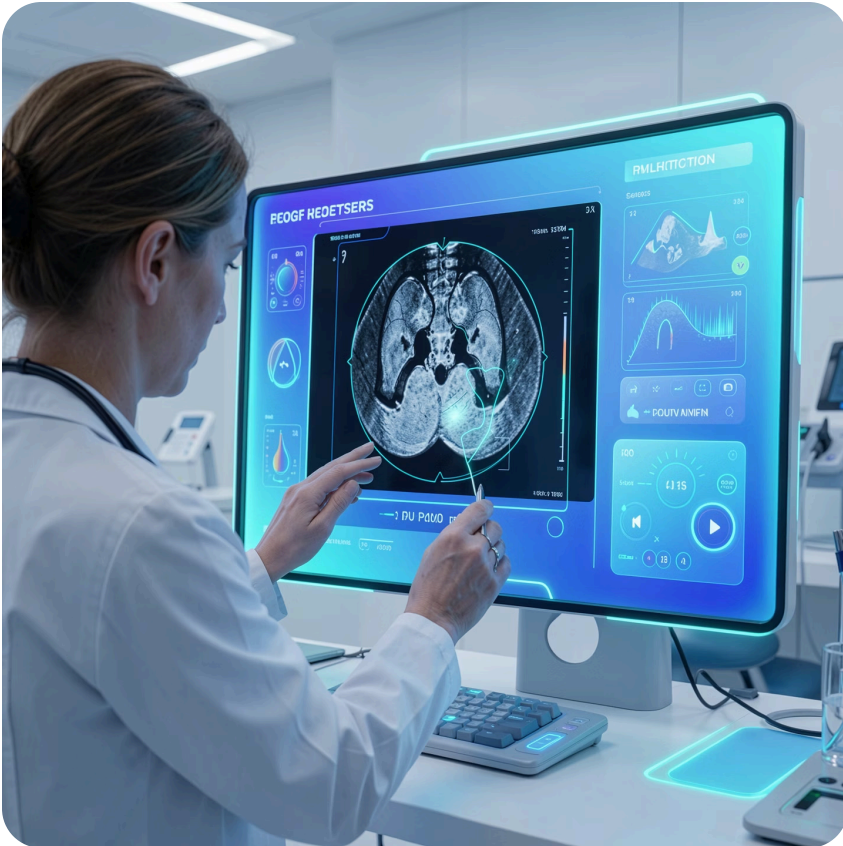


Tumor Sight: AI-Driven Breast Cancer Detection, Segmentation, and Classification



A Three stage Deep Learning Framework for Breast Cancer Diagnosis: YOLO26-Based Detection, U-Net Segmentation, and ResNet18 Classification

Authors

Matricola.

- | | |
|----------------------------|--------|
| 1. Amanuel Abebayehu Belay | 948422 |
| 2. Nigus Bezabh Haile | 947129 |
| 3. Mulat Yazew Chekol | 924343 |

Table of Contents

1	Clinical Problem & Motivation
2	Dataset & Augmentation
3	Model Architecture Overview
4	Stage 1: Object Detection
5	Stage 2: Segmentation
6	Stage 3: Classification
7	Regulatory Framework
8	Limitations & Future Work

Clinical Problem & Motivation

The Challenge

- Ultrasound interpretation is subjective and varies between radiologists
- High inter-observer variability leads to inconsistent diagnoses
- Diagnosis delay

The Need

- AI-assisted tools to support clinical decision-making
- Improved detection in dense breast tissue
- Fewer false positives and false negatives
- Reduction in variability between radiologists' interpretations

Our Objective

- Improve diagnostic accuracy and reduce human error
- Minimize inter-observer variability among radiologists
- Provide consistent and standardized interpretation of scans
- Assist clinicians in faster and more confident decision-making
- Enhance overall patient outcomes through timely diagnosis

Dataset & Data Augmentation

BUSI Dataset

The **Breast Ultrasound Image (BUSI)** dataset serves as our primary training resource, containing annotated ultrasound scans with expert-labeled tumor boundaries.

Additional Augmentation

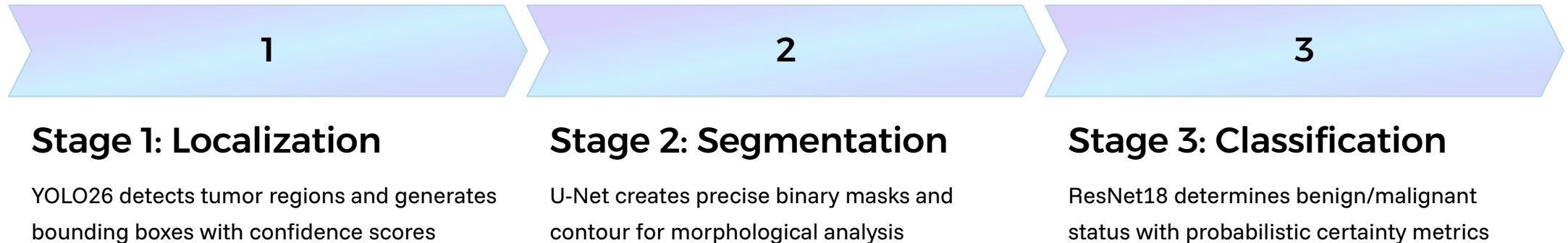
- Random rotations and flips for orientation invariance
- Brightness and contrast adjustments
- Scale and Shift

Dataset Characteristics

- Total 647 annotated ultrasound images
- Pixel-level segmentation masks for tumor boundaries
- Bounding box annotations for localization
- Classification labels

Overview: A 3-Stage Diagnostic Pipeline

Our AI model processes ultrasound images through a cascaded architecture designed to mimic the clinical decision-making workflow. Each stage builds upon the previous one, progressively refining the analysis from initial detection to definitive classification.



The modular design allows for independent optimization of each stage and facilitates interpretability through intermediate outputs.

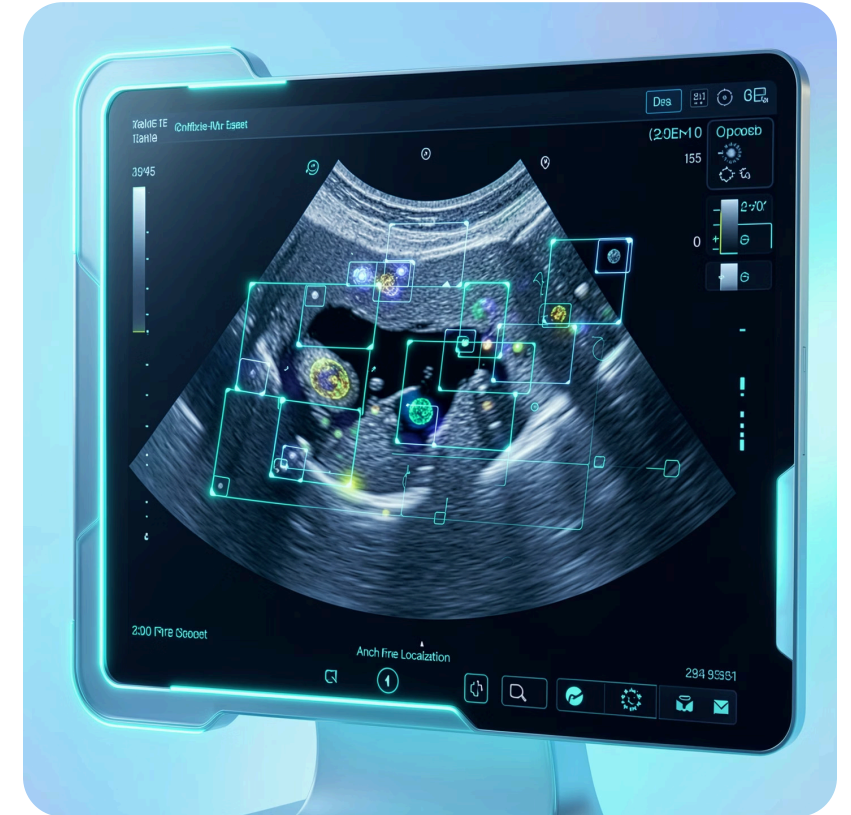
Stage 1: Object Detection with YOLO26

Implementation Details

Our object detection module employs YOLO26, a state-of-the-art single-stage detector that processes images in a single forward pass. The model uses the pre-trained **best.pt** weights, fine-tuned on annotated breast ultrasound datasets.

Key Features

- Multi-scale detection handles tumors of varying sizes
- Confidence-based Region of Interest (ROI) generation
- Real-time inference compatible with clinical workflows
- Bounding box area percentage relative to full image



The YOLO26 architecture employs a compound scaling method that balances network depth, width, and resolution. Its detection head uses anchor-free points with predicted offsets, reducing hyperparameter sensitivity while improving localization accuracy for irregularly shaped tumors.

Performance Analysis on YOLO26

79.43%

Mean Average Precision
(mAP)

IoU threshold of 0.5 across all tumor
classes

86.15%

Recall Rate

Sensitivity to detect all annotated tumor
regions

86.15%

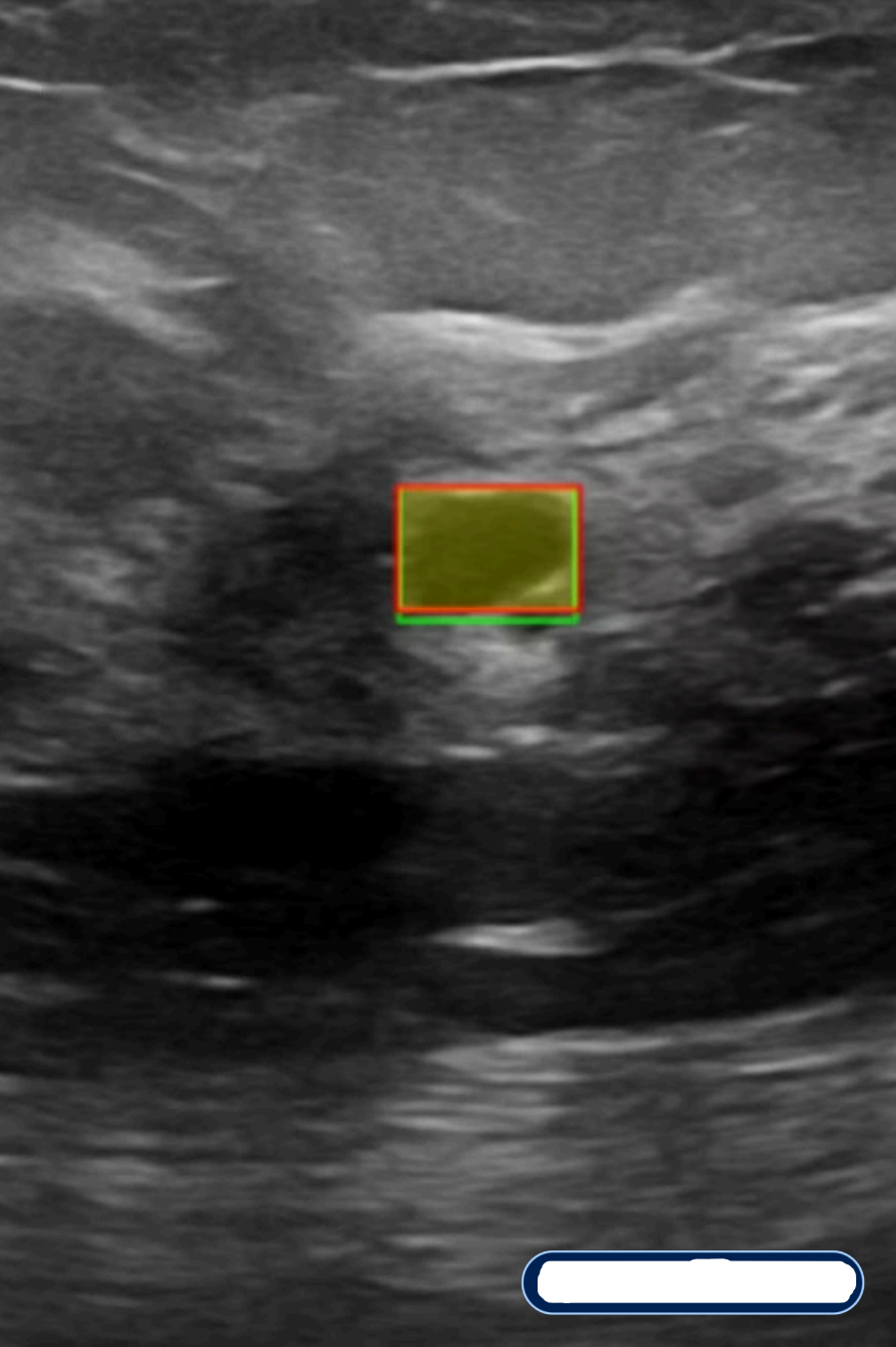
Precision

Proportion of detections that are true
positives

179.6ms

Inference Time

Average processing time per ultrasound
image



Stage 2: Segmentation with U-Net

U-Net architecture employs an encoder-decoder structure with skip connections, making it particularly effective for medical image segmentation tasks where precise boundary delineation is critical.

1

Encoder Path

Successive convolutional layers downsample the image, extracting increasingly abstract features while reducing spatial dimensions

2

Bottleneck

Compresses learned features into a compact representation that captures essential tumor characteristics

3

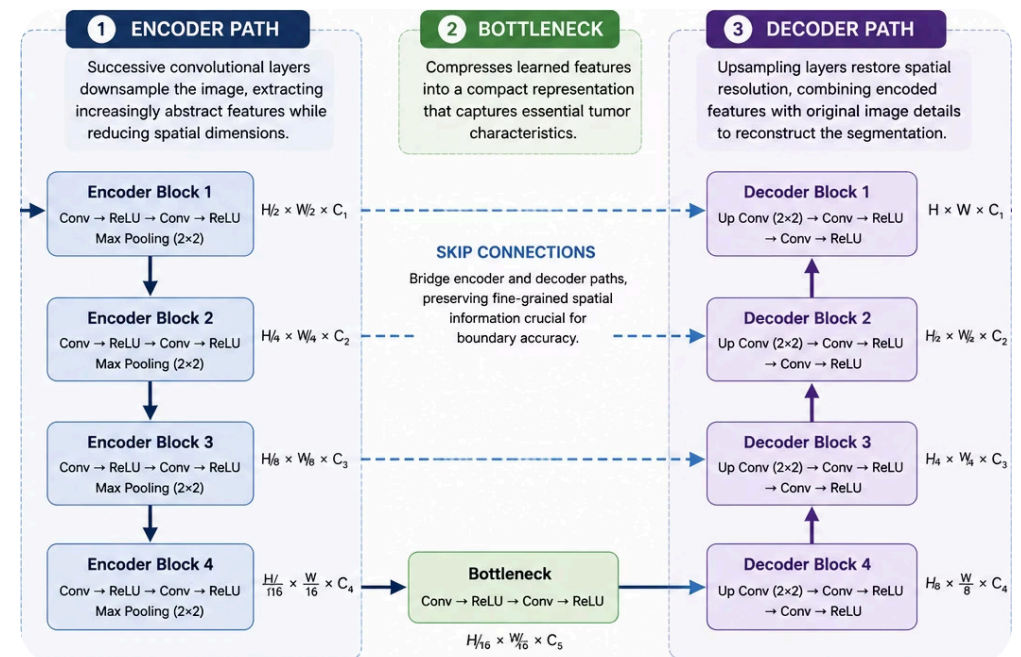
Decoder Path

Upsampling layers restore spatial resolution, combining encoded features with original image details

4

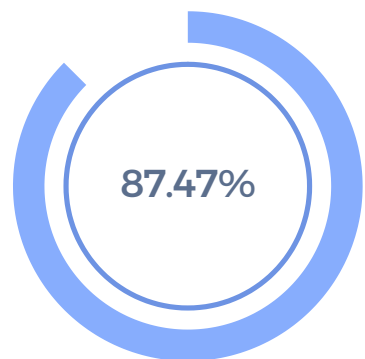
Skip Connections

Bridge encoder and decoder paths, preserving fine-grained spatial information crucial for boundary accuracy

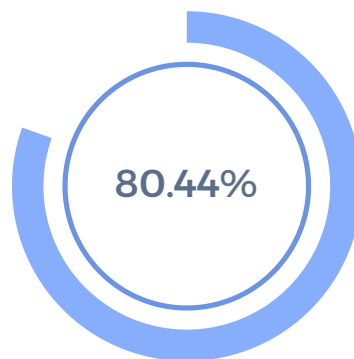


The skip connections in U-Net are particularly valuable for medical imaging because they allow the decoder to access high-resolution features from the encoder, enabling precise pixel-level predictions.

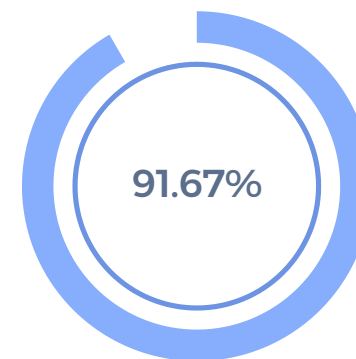
Performance Analysis on U-NET



Dice Coefficient



IoU Score

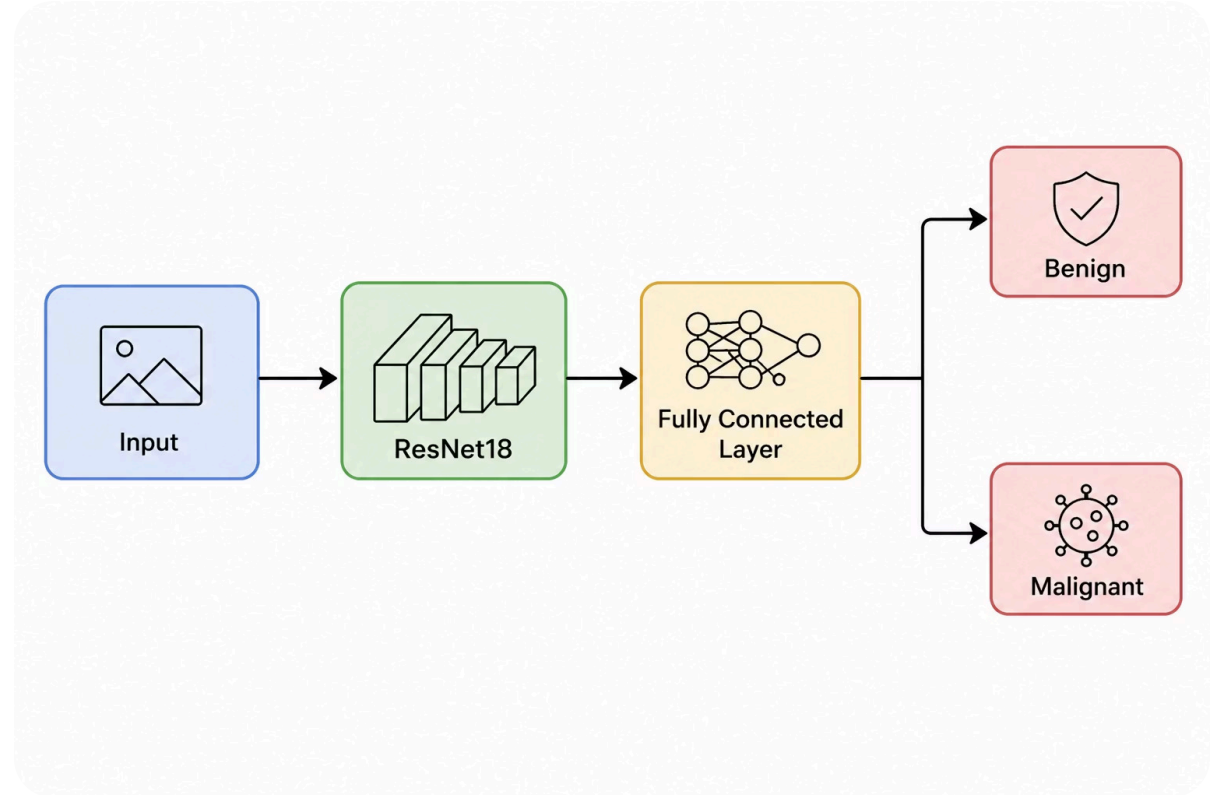


Recall

Stage 3: Classification with ResNet18

The model outputs binary decisions, either benign or malignant.

Training employed class-balanced sampling and focal loss to address dataset imbalance, achieving balanced sensitivity-specificity trade-offs suitable for screening applications.



Performance Analysis on ResNet18

ROC curve analysis reveals strong discriminative capability with Area Under Curve (AUC) of 0.96, indicating excellent separation between benign and malignant probability distributions. The curve demonstrates appropriate threshold placement, with Youden index identifying 0.45 as optimal cutoff point.

Accuracy

90.82%

Sensitivity

87.50%

Specificity

92.42%

ROC AUC

95.98%

Time

2.04 ms / img

True	benign	61	5
	malignant	4	28
		benign	malignant
		Predicted	

Regulatory Framework: EU AI Act & Medical Device Regulation



Medical Device Regulation 2017/745

EU framework governing medical devices, including software as medical device (SaMD). Classifies AI systems based on risk levels and intended clinical use.



AI Act

High-risk AI model for medical purposes require rigorous conformity assessment, technical documentation, and ongoing performance monitoring.



Documentation Requirements

Comprehensive technical files, clinical evaluation reports, risk management documentation, and quality management systems.

Key Regulatory Considerations

- **Intended Use Definition:** Clearly specify the clinical purpose and target patient population
- **Performance Evaluation:** Demonstrate analytical and clinical validity through appropriate datasets
- **Transparency:** Provide interpretable outputs and uncertainty quantification for clinical decision support
- **Robustness:** Ensure consistent performance across diverse clinical settings and patient demographics
- **Post-Market Surveillance:** Implement continuous monitoring and adverse event reporting systems

Limitations & Future Work

Current Limitations

- May not capture full real-world variability in clinical data
- Limited size of the training dataset
- Risk of overfitting and reduced robustness in underrepresented class
- Performance variation across different ultrasound vendors and devices
- Limited trust and interpretability for clinical use

Future Research Directions

DICOM/PACS Integration

Real-time image retrieval from hospital PACS systems using DICOM protocols, eliminating manual file transfers

Multi-Modal Analysis

Fusion of ultrasound with mammography and MRI data through late-stage neural network concatenation

Optimization & Expansion

Improve real-time speed and extend use to broader indications.

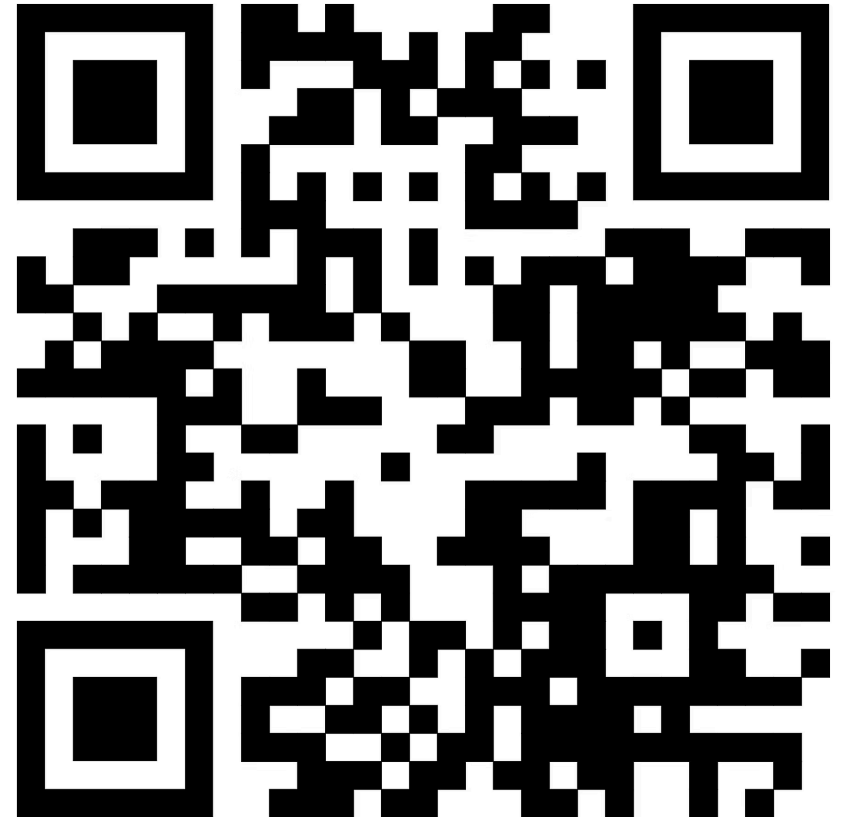
Risk level Analysis

Integrate a standard to map model predictions into clinically accepted risk categories.

Thank You

We appreciate your time, attention, and engagement.

It's been a pleasure sharing this with you.



Scan to test